Instructions for homework submission

1. Complete two parts in this homework:
    a. Math questions: Include your solution in LATEX document. Show your work. Submission with embedded photos of handwritten work will not be accepted.
    b. Programming questions: Complete the given skeleton Python code. **For questions requiring visualization, analysis, and discussion, please include your solution in the same LATEX document.**
2. Submit your work to Gradescope including:
    a. A PDF document for written parts: `FirstName_LastName_HW4.pdf`. LATEX source code is not required.
    b. A completed Python code: `FirstName_LastName_HW4.py`. *(Note: You may only use standard library (os, pickle, re, etc.), matplotlib, and library explicitly defined in given skeleton code.)*
    c. A pretrained model file: `model.pt`
    d. **There are two separate submission portals on Gradescope**: one for code and one for the report.
    e. Please assign your answer in PDF report to its corresponding question when submitting to Gradescope.
    f. Failure to follow submission instruction will result in loss of marks.
3. This is an individual assignment. While you may consult class mates and other resources, your code should be your own and your write up your own. Please cite any such external help in your write up.
4. Start early!
5. Total: 100 points
6. You should have already applied for and received access to the Grace Cluster from HPRC in HW1. Use the Grace Cluster if model training is taking too long or if you need additional computational resources. Refer to HW1 for more details.

This project utilizes a subset of the SUN397 database[1][2]. We acknowledge and appreciate their efforts in collecting and maintaining this data.

[1] Xiao, Jianxiong, et al. "Sun database: Large-scale scene recognition from abbey to zoo." 2010 IEEE computer society conference on computer vision and pattern recognition. IEEE, 2010.

[2] Xiao, Jianxiong, et al. "Sun database: Exploring a large collection of scene categories." International Journal of Computer Vision 119 (2016): 3-22.

1 Math Question: Convolution Operation

Perform convolution using the following matrix and filter, show your steps, and report your results.

Input matrix (7 by 7):

$$\begin{vmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 4 & 1 & 0 & 0 \\ 0 & 3 & 1 & 1 & 0 & 1 & 0 \\ 0 & 2 & 4 & 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 5 & 2 & 2 & 0 \\ 0 & 0 & 1 & 3 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{vmatrix}$$

Note: The outer ring of zeros represents zero-padding. The interior 5 by 5 region contains the active data.

Filter (3 by 3):

$$\begin{vmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{vmatrix}$$

Apply the convolution operation for all the following settings respectively and report your results. Your max pooling parameters should always be Pool size: 3 by 3; Stride: 2.

1.1 Operation: Without max pooling and with a convolution stride of 1.

Calculate and report:

- The dimensions of the output matrix
- The complete output matrix

Show your work.

1.2 Operation: Without max pooling and with a convolution stride of 2.

Calculate and report:

- The dimensions of the output matrix
- The complete output matrix

Show your work.

1.3 Operation: With max pooling and with a convolution stride of 1.

Calculate and report:

- The dimensions of the output matrix
- The complete output matrix

Show your work.

2 Programming Question: Image Classification using CNN

In this homework, you will implement two primary components:

1. A PyTorch Dataset class, SUN397Dataset , designed to load images from a given folder struc ture. You are asked to implement the SUN397Dataset class such that:
    a. It inherits from torch.utils.data.Dataset.
    b. It performs necessary transformation (i.e. normalization) and assigns an integer label (starting at `0`) based on the class name (ordered alphabetically). For normalization, you need to compute the mean/std before cropping or processing your input. The submitted mean/std is expected to be rounded to 4 decimal points. *(Note: normalization for test data will be handled by autograder. You can verify your computed mean/std is correct via testcase 2.2.)*
    c. Its constructor accepts a single path to your data folder, for example: `./data/train`.
    d. The `__getitem__` method correctly returns a tuple (image tensor, label tensor).
    e. The `__len__` method returns the total number of samples in your dataset.
2. A Convolutional Neural Network (CNN) model, to classify images into the appropriate classes. You have to use at least **ONE** convolutional layer (i.e., `nn.Conv2d`). Your CNN class is expected to work with image input of size 224x224.

You will then train your CNN on a subset of the SUN397 dataset, save your best trained model to model.pt and aim for 75% classification accuracy on our reserved test set.

2.1 Best Practices & Tips

None of these are mandatory, but you might find them helpful along the way!

1. Check your dataset: Verify that iterating over your SUN397Dataset returns (image, label) pairs with the expected shapes. Debugging issues here first will help you avoid errors in training. You can create a validation split to better understand your model behavior.
2. Network architecture: Perform a parameter count to identify which components of your network hold the most parameters, and analyze whether they are contributing effectively to the learning process.
3. Hyperparameters: Tweak schedulers, optimizers, and other hyperparameters to improve your model.
4. Data augmentation: You may incorporate additional transforms (random cropping, flipping, etc.) during training to generalize better.
5. Argparse: An example of an argument parser is provided to you, aiming to improve your hyperparameter tuning efficiency.

2.2 Important Constraints

- You are NOT allowed to use pretrained model from internet. Your submitted model weights need to be learned from scratch.
- DO NOT attempt to manipulate your training data in a way that gives you an unfair advantage.
- Failure to follow these constraints will disqualify your submission.

3 Use of AI and External Resources

The purpose of this section is to cite the external resources you used in creating your homework. In particular, if you use any generative AI tools to help generate your code you should then explain:

1. What were the prompts you provided the AI tools?
2. What mistakes did you catch the AI making?
3. What did you do to validate the correctness of the AI generated portions, aside from relying on Gradescope Autograder scoring?